

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

$$\begin{aligned} a. \quad SST &= \sum y_i^2 - N\bar{y}^2 \\ &= 7825.94 - 20(19.21)^2 \\ &= 445.458 \\ R^2 &= \frac{SSR}{SST} = \frac{375.47}{445.458} \approx 0.8429 \end{aligned}$$

$$\begin{aligned} b. \quad SSE &= SST(1 - R^2) \\ &= 725.94(1 - 0.7911) \\ &\approx 151.6489 \\ \hat{\sigma}^2 &= \frac{SSE}{N-2} = \frac{151.6489}{20-2} \approx 8.4249 \end{aligned}$$

$$\begin{aligned} c. \quad SST &= \sum (y_i - \bar{y})^2 = 631.63 \\ SSE &= \sum \hat{e}_i^2 = 182.85 \\ R^2 &= 1 - \frac{SSE}{SST} \\ &= 1 - \frac{182.85}{631.63} \\ &= 1 - 0.2895 \\ &\approx 0.7105 \end{aligned}$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- a. the predicted value of y for $x_0 = 4$.

$$\hat{y}_0 = 1.2 + 0.8(4) = 4.4$$

- b. the $se(f)$ corresponding to part (a).

$$\begin{aligned} Var(f) &= \hat{\sigma}^2 \left(1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right) \\ &= 1.2 \left(1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right) \\ &= 2.52 \end{aligned}$$

$$se(f) = \sqrt{2.52} \approx 1.5875$$

- c. a 95% prediction interval for y given $x_0 = 4$.

$$\therefore t_{0.95, 3} = 3.182$$

$$\begin{aligned}\hat{y}_0 \pm t_c \cdot se(f) &= 4.4 \pm 3.182 \cdot 1.5875 \\ &= 4.4 \pm 5.0514\end{aligned}$$

$$\therefore 95\% \text{ P.I.} = [-0.6514, 9.4514]$$

- d. a 99% prediction interval for y given $x_0 = 4$.

$$\therefore t_{0.99, 3} = 5.841$$

$$\begin{aligned}\hat{y}_0 \pm t_c \cdot se(f) &= 4.4 \pm 5.841 \cdot 1.5875 \\ &= 4.4 \pm 9.2726\end{aligned}$$

$$\therefore 99\% \text{ P.I.} = [-4.8726, 13.6726]$$

- e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

$$\hat{y}_0 = 1.2 + 0.8 \cdot 1 = 2$$

$$\begin{aligned}\text{Var}(f) &= 1.2 \left(1 + \frac{1}{5} + \frac{(1-1)^2}{15} \right) \\ &= 1.44\end{aligned}$$

$$se(f) = \sqrt{1.44} = 1.2$$

$$\begin{aligned}\hat{y}_0 \pm t_c \cdot se(f) &= 2 \pm 3.182 \cdot 1.2 \\ &= 2 \pm 3.8184\end{aligned}$$

$$95\% \text{ P.I.} = [-1.8184, 5.8184]$$

$$(c) : 9.4514 - (-0.6514) = 10.1028$$

$$(e) : 5.8184 - (-1.8184) = 7.6368$$

width of interval in (e) is narrower than the interval in (c)
 when x_0 is closer to $\bar{x}$, $\frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}$ becomes smaller,
 making standard error becomes smaller

$\Rightarrow$ The prediction interval in (e) is narrower.